
Vehicle Market Analytics System

Database Setup — DDL Statements

Milestone 4 — CREATE TABLE, Constraints and Keys

Prepared By	Muddassir Khan, Salman Khan
Instructor	Ali Hassan
Course	Introduction to Database Systems
Institution	Institute of Management Sciences, Peshawar
Submission	May 2026

Schema Summary

12	17	3NF
Tables	FK Relations	Normal Form

Contents

1	Introduction	2
2	Database and Table Definitions	2
2.1	Database Setup	2
2.2	locations	2
2.3	categories	2
2.4	users	2
2.5	admins	2
2.6	sellers	3
2.7	vehicles	3
2.8	vehicle_images	4
2.9	favorites	4
2.10	comparisons	4
2.11	inquiries	4
2.12	reviews	5
2.13	search_logs	5
3	Constraint Summary	6

1. Introduction

This document contains the complete DDL for **pakwheels_db**, the 3NF-normalised relational schema for the Vehicle Market Analytics System. All 12 tables are defined with primary keys, foreign key references, NOT NULL constraints, and UNIQUE or CHECK constraints where applicable. Tables are defined in dependency order so that every referenced parent table exists before its child tables.

2. Database and Table Definitions

2.1 Database Setup

```
CREATE DATABASE IF NOT EXISTS pakwheels_db
  CHARACTER SET utf8mb4
  COLLATE utf8mb4_unicode_ci;

USE pakwheels_db;
```

2.2 locations

```
CREATE TABLE locations (
  location_id INT UNSIGNED NOT NULL AUTO_INCREMENT,
  city        VARCHAR(100) NOT NULL,
  state       VARCHAR(100) NOT NULL,
  country     VARCHAR(80)  NOT NULL DEFAULT 'Pakistan',
  PRIMARY KEY (location_id)
);
```

2.3 categories

```
CREATE TABLE categories (
  category_id INT UNSIGNED NOT NULL AUTO_INCREMENT,
  category_name VARCHAR(80) NOT NULL,
  description  TEXT,
  PRIMARY KEY (category_id),
  UNIQUE KEY  uq_cat_name (category_name)
);
```

2.4 users

```
CREATE TABLE users (
  user_id INT UNSIGNED NOT NULL AUTO_INCREMENT,
  name     VARCHAR(120) NOT NULL,
  email    VARCHAR(180) NOT NULL,
  password_hash VARCHAR(255) NOT NULL,
  phone    VARCHAR(20),
  role     ENUM('admin', 'seller', 'buyer')
           NOT NULL DEFAULT 'buyer',
  is_active TINYINT(1) NOT NULL DEFAULT 1,
  PRIMARY KEY (user_id),
  UNIQUE KEY  uq_email (email)
);
```

2.5 admins

```

CREATE TABLE admins (
  admin_id      INT UNSIGNED NOT NULL AUTO_INCREMENT,
  user_id       INT UNSIGNED NOT NULL,
  permissions   JSON          NOT NULL DEFAULT ('{}'),
  PRIMARY KEY   (admin_id),
  UNIQUE KEY    uq_admin_user (user_id),
  FOREIGN KEY   (user_id) REFERENCES users(user_id)
    ON DELETE CASCADE ON UPDATE CASCADE
);

```

2.6 sellers

```

CREATE TABLE sellers (
  seller_id      INT UNSIGNED NOT NULL AUTO_INCREMENT,
  user_id        INT UNSIGNED NOT NULL,
  location_id    INT UNSIGNED,
  business_name  VARCHAR(150),
  rating         DECIMAL(3,2) NOT NULL DEFAULT 0.00,
  seller_type    ENUM('individual','dealer','certified_dealer')
    NOT NULL DEFAULT 'individual',
  verified_status TINYINT(1) NOT NULL DEFAULT 0,
  PRIMARY KEY    (seller_id),
  UNIQUE KEY     uq_seller_user (user_id),
  FOREIGN KEY     (user_id) REFERENCES users(user_id)
    ON DELETE CASCADE ON UPDATE CASCADE,
  FOREIGN KEY     (location_id) REFERENCES locations(location_id)
    ON DELETE SET NULL ON UPDATE CASCADE
);

```

2.7 vehicles

```

CREATE TABLE vehicles (
  vehicle_id     INT UNSIGNED NOT NULL AUTO_INCREMENT,
  brand           VARCHAR(80) NOT NULL,
  model          VARCHAR(120) NOT NULL,
  year           YEAR          NOT NULL,
  price          BIGINT UNSIGNED NOT NULL,
  mileage        INT UNSIGNED NOT NULL DEFAULT 0,
  fuel_type      ENUM('Petrol','Diesel','Hybrid',
    'Electric','CNG','LPG') NOT NULL,
  transmission   ENUM('Automatic','Manual') NOT NULL,
  body_type      VARCHAR(60) NOT NULL,
  engine_cc      INT UNSIGNED,
  color          VARCHAR(60),
  status         ENUM('available','sold','featured')
    NOT NULL DEFAULT 'available',
  category_id    INT UNSIGNED,
  seller_id      INT UNSIGNED NOT NULL,
  location_id    INT UNSIGNED,
  PRIMARY KEY    (vehicle_id),
  CONSTRAINT chk_price CHECK (price > 0),
  CONSTRAINT chk_year  CHECK (year > 1980),
  FOREIGN KEY     (category_id) REFERENCES categories(category_id)
    ON DELETE SET NULL ON UPDATE CASCADE,
  FOREIGN KEY     (seller_id) REFERENCES sellers(seller_id)

```

```
    ON DELETE CASCADE ON UPDATE CASCADE,  
    FOREIGN KEY (location_id) REFERENCES locations(location_id)  
    ON DELETE SET NULL ON UPDATE CASCADE  
);
```

2.8 vehicle_images

```
CREATE TABLE vehicle_images (  
    image_id    INT UNSIGNED NOT NULL AUTO_INCREMENT,  
    vehicle_id  INT UNSIGNED NOT NULL,  
    image_url   VARCHAR(400) NOT NULL,  
    is_primary  TINYINT(1)   NOT NULL DEFAULT 0,  
    PRIMARY KEY (image_id),  
    FOREIGN KEY (vehicle_id) REFERENCES vehicles(vehicle_id)  
    ON DELETE CASCADE ON UPDATE CASCADE  
);
```

2.9 favorites

```
CREATE TABLE favorites (  
    favorite_id INT UNSIGNED NOT NULL AUTO_INCREMENT,  
    user_id     INT UNSIGNED NOT NULL,  
    vehicle_id  INT UNSIGNED NOT NULL,  
    PRIMARY KEY (favorite_id),  
    UNIQUE KEY  uq_fav (user_id, vehicle_id),  
    FOREIGN KEY (user_id) REFERENCES users(user_id)  
    ON DELETE CASCADE ON UPDATE CASCADE,  
    FOREIGN KEY (vehicle_id) REFERENCES vehicles(vehicle_id)  
    ON DELETE CASCADE ON UPDATE CASCADE  
);
```

2.10 comparisons

```
CREATE TABLE comparisons (  
    comparison_id INT UNSIGNED NOT NULL AUTO_INCREMENT,  
    user_id       INT UNSIGNED NOT NULL,  
    vehicle_id_1  INT UNSIGNED NOT NULL,  
    vehicle_id_2  INT UNSIGNED NOT NULL,  
    vehicle_id_3  INT UNSIGNED,  
    PRIMARY KEY   (comparison_id),  
    FOREIGN KEY   (user_id) REFERENCES users(user_id)  
    ON DELETE CASCADE ON UPDATE CASCADE,  
    FOREIGN KEY   (vehicle_id_1) REFERENCES vehicles(vehicle_id)  
    ON DELETE CASCADE ON UPDATE CASCADE,  
    FOREIGN KEY   (vehicle_id_2) REFERENCES vehicles(vehicle_id)  
    ON DELETE CASCADE ON UPDATE CASCADE,  
    FOREIGN KEY   (vehicle_id_3) REFERENCES vehicles(vehicle_id)  
    ON DELETE SET NULL ON UPDATE CASCADE  
);
```

2.11 inquiries

```
CREATE TABLE inquiries (  
    inquiry_id INT UNSIGNED NOT NULL AUTO_INCREMENT,  
    user_id    INT UNSIGNED,  
    vehicle_id INT UNSIGNED NOT NULL,
```

```

seller_id  INT UNSIGNED  NOT NULL,
message    TEXT          NOT NULL,
status     ENUM('new', 'read', 'replied', 'closed')
           NOT NULL DEFAULT 'new',

PRIMARY KEY (inquiry_id),
FOREIGN KEY (user_id) REFERENCES users(user_id)
  ON DELETE SET NULL ON UPDATE CASCADE,
FOREIGN KEY (vehicle_id) REFERENCES vehicles(vehicle_id)
  ON DELETE CASCADE ON UPDATE CASCADE,
FOREIGN KEY (seller_id) REFERENCES sellers(seller_id)
  ON DELETE CASCADE ON UPDATE CASCADE
);

```

2.12 reviews

```

CREATE TABLE reviews (
  review_id  INT UNSIGNED  NOT NULL AUTO_INCREMENT,
  reviewer_id INT UNSIGNED  NOT NULL,
  seller_id  INT UNSIGNED  NOT NULL,
  rating     TINYINT UNSIGNED NOT NULL,
  comment    TEXT,
  PRIMARY KEY (review_id),
  CONSTRAINT chk_rating CHECK (rating BETWEEN 1 AND 5),
  FOREIGN KEY (reviewer_id) REFERENCES users(user_id)
    ON DELETE CASCADE ON UPDATE CASCADE,
  FOREIGN KEY (seller_id) REFERENCES sellers(seller_id)
    ON DELETE CASCADE ON UPDATE CASCADE
);

```

2.13 search_logs

```

CREATE TABLE search_logs (
  log_id      INT UNSIGNED  NOT NULL AUTO_INCREMENT,
  user_id     INT UNSIGNED,
  query_text  VARCHAR(300),
  results_count INT UNSIGNED,
  searched_at TIMESTAMP     NOT NULL DEFAULT CURRENT_TIMESTAMP,
  PRIMARY KEY (log_id),
  FOREIGN KEY (user_id) REFERENCES users(user_id)
    ON DELETE SET NULL ON UPDATE CASCADE
);

```

3. Constraint Summary

Table	Constraint	Purpose
categories	UNIQUE uq_cat_name	No duplicate category names.
users	UNIQUE uq_email	Each email is unique across all accounts.
admins	UNIQUE uq_admin_user	One admin record per user (1:1).
sellers	UNIQUE uq_seller_user	One seller profile per user (1:1).
vehicles	CHECK chk_price	Price must be greater than zero.
vehicles	CHECK chk_year	Manufacturing year must be after 1980.
favorites	UNIQUE uq_fav	A user cannot save the same vehicle twice.
reviews	CHECK chk_rating	Rating must be between 1 and 5.

Table 1: Key constraints applied across the schema